

Product Requirements Document

MyCCAngel

Status: Active draft — reflects both shipped functionality and target-state product vision

Owner: Founding Product Lead

Last updated: June 2026

Status legend used throughout this document:

- **Shipped** — built and functional ● **In Development** — actively being built ○ **Planned** — specified, not started

1. Problem Statement

People living with Inflammatory Bowel Disease (IBD) face challenges far beyond clinical management of their condition. Reliable information is fragmented across dozens of sources. Trustworthy peer communities are hard to find. Severe symptoms — including bathroom urgency occurring dozens of times daily — disrupt work, school, travel, and relationships, often without any centralized place to find practical, day-to-day support.

There is currently no single platform that combines verified peer community, centralized healthcare resources, and personalized AI-driven guidance specifically for IBD patients.

2. Goals

Goal	Description
Trust	Ensure the community consists exclusively of verified IBD patients
Connection	Reduce isolation by connecting patients with peers who share similar experiences
Personalization	Surface the right people, content, and resources to each patient from day one
Practical support	Help patients navigate daily-life challenges (diet, bathroom access, symptom management)
Centralization	Bring fragmented healthcare information into one trusted destination

3. Non-Goals

- MyCCAngel does not provide medical diagnosis or treatment recommendations.
- MyCCAngel does not replace a patient's relationship with their treating physician.
- The AI chatbot is explicitly scoped to answer general informational questions and guide users to resources — not to give personalized medical advice.

4. Target Users & Personas

Primary Persona: Diagnosed IBD Patient

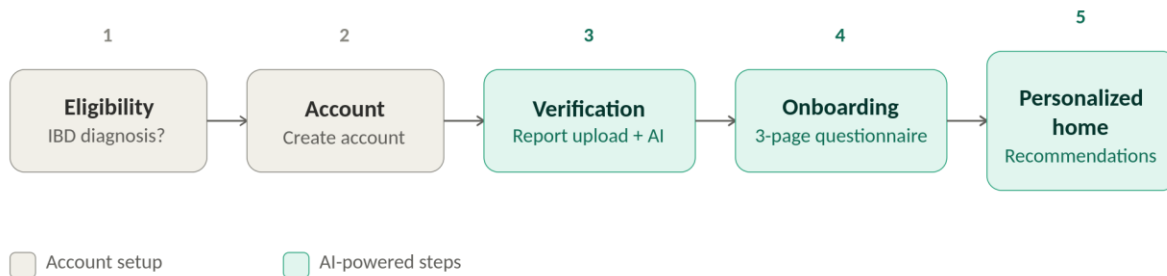
- Diagnosed with Crohn's Disease or Ulcerative Colitis
- Seeking community, practical daily-life support, and trustworthy information
- Willing to verify diagnosis via medical report to access a trusted, patient-only space

Secondary Stakeholders

- Healthcare professionals (doctors, dietitians) listed in the resource directory
- Partner organizations contributing educational content

5. User Journey Overview

1. Prospective user answers an eligibility question (diagnosed with IBD: yes/no)
2. Creates an account
3. Uploads a medical report for AI-based verification
4. Completes a 3-page onboarding questionnaire
5. Lands on a personalized home experience with recommended people, content, and resources



6. Feature Requirements

6.1 Eligibility & Medical Verification • Shipped

Requirement: Before full registration, users confirm an IBD diagnosis. Users then upload a medical report, which is processed through an AI verification workflow before platform access is granted.

Functional details:

- OCR-based text extraction from uploaded reports (PyPDF2, Pytesseract)
- Text preprocessing and cleaning
- SVM classification model distinguishing IBD vs. non-IBD reports
- Trained and evaluated on a manually labeled dataset of 200 reports; ~91% accuracy on a held-out test split

Acceptance criteria:

- A user without a qualifying medical report cannot complete registration
- Verification decision returned without manual review in the standard case

6.2 Onboarding Questionnaire • Shipped

Requirement: A 3-page questionnaire collects the data needed to power personalization, immediately after verification.

Page 1 — Understanding the Patient

- Year of diagnosis; IBD type (Crohn's / Ulcerative Colitis)
- Disease status (remission / flare-up / awaiting clinical evidence / symptomatic) with status-dependent follow-ups
- Symptom severity: toilet frequency, presence of blood/mucus in stool
- Daily-life impact: overall severity and most-affected areas (work/school, social, physical health, mental health, diet, sleep, relationships, etc.)

Page 2 — Support Needs

- Toilet locator interest
- IBD-friendly diet/recipe interest
- IBD-friendly restaurant interest
- Dietitian referral interest
- Diagnosis-related support interest

Page 3 — Consent & Research

- Research participation consent
- Consent for use of data in recommendations
- Consent for anonymized data analysis
- Terms & conditions agreement

Design rationale: Every question maps to a specific downstream use (verification, recommendation, or research) rather than serving as generic intake — this should be preserved in any future iteration.

6.3 Recommendation Engine • Shipped

Requirement: Personalize the new-user experience using onboarding data, before the user has taken any in-platform action.

Inputs used: diagnosis year, disease type/status, symptom severity, toilet frequency, blood/mucus indicators, daily-life impact areas, support needs, geography, and other profile attributes.

Outputs:

- Recommended people (similar disease experience, similar symptoms, patients in remission)
- Recommended resources (educational articles, healthcare organizations, research opportunities)
- Recommended healthcare support (doctors, dietitians, IBD-friendly restaurants, toilet locations)
- Recommended communities (relevant groups, discussions, events)

Acceptance criteria:

- A new user receives at least one recommendation in each category within their first session
- Recommendations update as profile/survey data is added or changed

6.4 AI Healthcare Assistant (Chatbot) • In Development

Requirement: A conversational assistant that answers common IBD questions, explains healthcare concepts in plain language, and directs users to relevant platform resources — without diagnosing or prescribing.

Current state: A custom NLP/ML chat pipeline is built and functional — it can hold a conversation. The remaining step is training on a curated, trusted IBD content set so responses are accurate and on-topic.

Remaining requirements to reach Shipped:

- Curate and structure a trusted IBD knowledge base (medical literature, partner-organization content, FAQ banks)
- Train/fine-tune the pipeline on this content
- Add guardrails preventing diagnostic or prescriptive responses
- Add escalation path to human/professional resources for out-of-scope queries
- QA conversation accuracy against a held-out question set before launch

6.5 Community & Profile • Shipped (core) • In Development (extended)

Shipped:

- User profiles (medical info: disease status, IBD type, years since diagnosis)
- Community posts, comments, reactions
- Connections and direct messaging, surfaced via AI-driven people recommendations

In Development / Planned:

- Group creation and management (roles, public/private/secret privacy tiers)
- Group chat, @mentions, file sharing in groups
- Message read receipts, typing indicators, message reactions
- Search within conversations

6.6 Resource Hub ○ Planned

Requirement: A centralized directory of healthcare-related information and support services.

Scope:

- Doctor directory
- Dietitian directory
- Partner organization listings
- Educational articles and IBD management resources
- Toilet locator, including crowdsourced location submissions from users (with a moderation/review path before a submission becomes publicly visible)
- IBD-friendly restaurant finder

Current state: Page layouts and visual design are complete (see Figma/frontend); directories are not yet populated with real data and are not connected to a live backend.

6.7 Symptom & Concern Support Center ○ Planned

Requirement: A homepage section where patients select a specific concern and receive tailored educational content, articles, community discussions, resource recommendations, and relevant healthcare professionals.

Concerns in scope (25 total): abdominal pain/cramping, blood in stool, mucus in stool, fatigue, diarrhea, constipation, weight loss/malnutrition, nausea/vomiting, loss of appetite, bloating/gas, urgency/frequency, incontinence, rectal pain/pressure, fever/chills, anemia, osteoporosis, skin rashes/lesions, joint pain/arthritis, eye problems, liver problems, kidney stones, osteopenia, anxiety/depression, sleep disturbances, sexual dysfunction.

Current state: Concept and content mapping defined; not yet built.

6.8 Events ○ Planned

Requirement: Community events, educational sessions, and awareness activities, listed and joinable within the platform.

Current state: Scoped, not yet built.

6.9 General Platform Requirements

Requirement	Status
Responsive across devices and major browsers	In Development
Multi-language content support	Planned
Payment processing (donations, premium features)	Planned
Automated email notifications (registration, password reset)	Planned
Data backup & restore	Planned

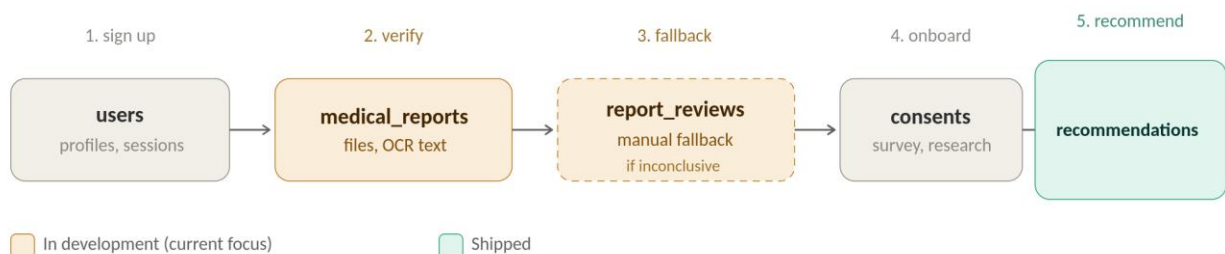
7. Backend Architecture

• In Development

The backend was in active development at the time of project pause, with priority given to the most complex flow: signup → eligibility check → medical report upload → AI verification → onboarding questionnaire → recommendation generation. This sequencing was deliberate — it's the highest-risk, highest-value flow, since every other feature depends on a verified, profiled user existing first.

Data model: A relational schema (PostgreSQL) was designed covering core identity (users, profiles, partner profiles, sessions, consents), medical verification (files, medical reports, and a dedicated report-review table for manual fallback when automated classification is inconclusive), social features (connections, recommendations, notifications), messaging (conversations, members, messages), content (posts, attachments, comments, reactions), the resource library (doctors, dietitians, organizations, articles, concerns), the toilet locator (including crowdsourced point-of-interest submissions with a moderation path), AI chat sessions/messages, donations, and audit logging.

Including a dedicated audit log and an explicit consent table at the schema level — rather than treating them as afterthoughts — reflects deliberate attention to the trust and compliance requirements of a healthcare product, not just its user-facing features.



Outstanding work: completing and integrating backend services for the resource hub, community/group features, and the symptom support center; connecting the trained chatbot; full QA across the signup chain end-to-end.

8. Success Metrics (Target, Post-Launch)

- Verification model precision/recall on production traffic, monitored against the 200-report validation baseline
- % of new users completing onboarding questionnaire
- % of new users engaging with at least one recommendation within first session
- Community engagement: posts/connections per active user per week
- Chatbot resolution rate (post-training): % of queries answered without escalation

9. Open Questions

- What is the minimum viable trusted-content dataset size for chatbot training?
- Should the resource hub launch with a curated initial dataset, or require partner organizations to onboard before launch?
- What is the data privacy/retention policy for uploaded medical reports post-verification?

10. Status Summary Table

Feature Area	Status
Medical Report Verification	Shipped
Onboarding Questionnaire	Shipped
Recommendation Engine	Shipped
Core Community & Profile	Shipped
AI Healthcare Assistant	In Development
Backend (signup/verification chain)	In Development
Responsive Design (full QA)	In Development
Extended Community (groups, chat features)	Planned
Resource Hub	Planned
Symptom & Concern Support Center	Planned
Events	Planned
Payments, Notifications, Multi-language	Planned